

Experiment No. 9

PHOTO CELL CHARACTERISTICS

Aim: To verify the inverse square law of light with the help of Selenium Photocell

Apparatus: Selenium Photocell, electric lamp (bulb), multimeters.

Theory:

A device used to convert light energy into electrical energy is called **Photo Electric Cell**. Photo Electric Cell is based on the phenomenon of Photo Electric Effect. Photo Electric Cells are of three types.

1. Photo-Emissive Cell.
2. Photo-Voltaic Cell.
3. Photo-Conductive Cell.

1. Photo –Emissive Cell:

There are two types of Photo-emissive cells

- A. Vacuum Type Cell
- B. Gas Filled Type Cell

It consists of two electrodes i.e. cathode (K) and anode (A). The cathode is in the form of semi-cylindrical plate coated with photo-sensitive material like sodium potassium or cesium i.e. alkali metals. To have large current, it is usually coated with antimony cesium alloy or combination of bismuth, silver, oxygen and cesium. The anode (A) is in the form of a straight wire made of nickel or platinum. The anode (A) faces the cathode (K). These electrodes are sealed in an evacuated glass or quartz bulb according to whether it is to be used with visible or ultra-violet light. As the current due to vacuum is small, so to increase the current, the bulb of the cell is filled with an inert gas like helium, neon, argon etc. At pressure of 1 mm of mercury. When Photo-Electrons flow from cathode to anode, they ionize the gas filled and hence the current gets modified. The main

drawback of this type of cell (i.e. gas filled cell) is that the photo-electric current does not vary linearly with intensity of the light.

Since there is no time lag between the incident light and the flow of electrons and hence current, therefore such a cell is used in television, photometry, fire alarm etc.

2. Photo-Voltaic Cell

Photo-Voltaic Cell is based on the principle of inner photo electric cell. This is called true cell because it generates e.m.f. without the application of any external potential difference but by only the light incident on it. It consists of a semi-conductor layer formed on the surface of the metal plate by either heat treatment or cathode sputting. A thin film of semi-transparent metal is coated over the semi-conductor. This film maintains the electrical contact with the semi-conductor and simultaneously allows the incident light to fall on the semi-conductor.

When light is incident on the semi-conductor, electrons are emitted which flow in a direction opposite to the light rays. If the circuit is completed between the surface transparent film and metal base through a low resistance galvanometer (G), the current can be measured. If the resistance of the circuit is very small, the current is proportional to the intensity of the incident light. The main advantage of this cell is that it requires no external voltage for its operation. This type of cell is widely used in photographic exposure meters, photometers and illumination meters etc.

3. Photo-Conductive Cell

Photo-Conductive cell is also based on the principle of inner Photo-Electric Effect. It consists of a thin film of semi-conductor like selenium or thallous sulphide place below a thin film of semi-transparent metal. This combination is placed over the block of iron. The iron base and the transparent metal film is connected through battery and resistance. When light falls on the cell, its resistance decreases and hence the current starts flowing in the external circuit.

If 'I' be the luminous intensity of an electric lamp and 'E' be the luminance at a point distance "d" from it, then according to the inverse square law;

$$E = I/d^2$$

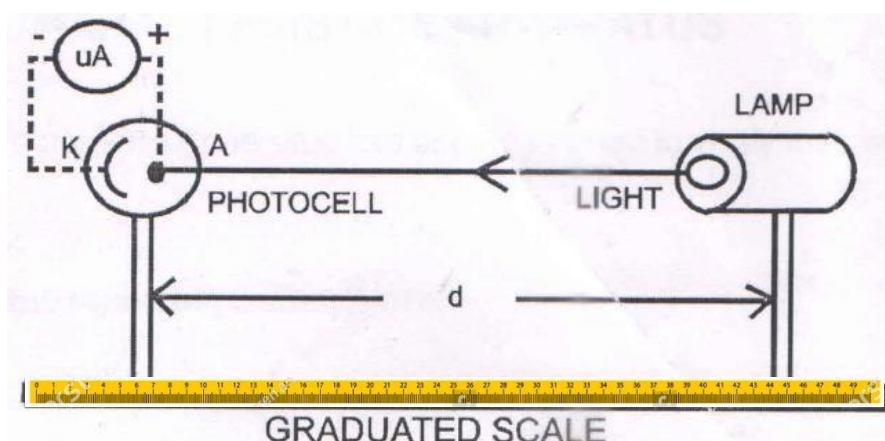
If light from the lamp be incident on the photovoltaic cell placed at a distance 'd' from it and if q be the corresponding deflection shown by the micrometer then,

$$q \propto \frac{I}{d^2} \rightarrow q = k \frac{I}{d^2}, \quad \rightarrow \quad qd^2 = \text{constant}$$

The deflection of micrometer can be measure in multimeter current 'i' in (mA)

Procedure

1. The experiment can be performed in the laboratory but it is always good to perform it in a dark room where stray light falling on the photocell can be avoided. In the dark room mount the various parts of the apparatus on optical bench provided with a $\frac{1}{2}$ meter scale. Make the other connections as shown in the figure given below.



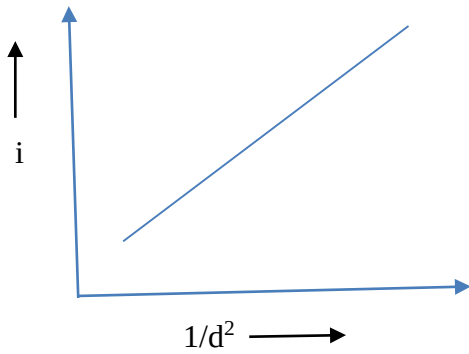
2. Light up the lamp and adjust it at a suitable distance from the photocell so that the micrometer and voltmeter indicates a reasonable deflection.
3. Change the distance of lamp from the photo voltaic cell and take a series of observations for the corresponding values of d and q .

Observation Table:

S. N.	d (cm)	I (mA)	$1/d^2$ (m^{-2})	$I \times d^2$
1				
2				
3				
.				
.				
.				
.				

Graph

Plot a graph between $1/d^2$ and i , taking $1/d^2$ on x-axis and i on y-axis



Result

The plotted graph is obtained as straight line, so inverse law is verified.

Precautions

1. Stray Light should be avoided.
2. The effect of the reflected light from the bench surface should be minimized.
3. Very sensitive micrometer should be used.